

Syllabus for the Exam of Junior Engineer (Civil) to be conducted by HPRCA

1. SUBJECTS AS PER ESSENTIAL QUALIFICATION:

- a) **CONSTRUCTION MATERIALS:** Classification of construction materials: Natural, Artificial, Special, Finishing, Recycled; Properties and uses of: -Stone, Timber, Lime, Sand, Coarse aggregate, Bitumen, Asphalt, Tar; Bricks: -Conventional bricks, Fly Ash bricks, Aerated Concrete blocks, Field tests and classification; Cement: - Manufacturing, Types and uses, Field Tests, Lab Test; Special and processed materials: -Waterproofing Materials, Thermal and Sound Insulation, Geopolymer Cement, POP, Paints and Varnishes, Industrial/Agro Waste Materials.
- b) **BASIC SURVEYING:** Principles and classification of surveying; Chain surveying: - Instruments, Ranging, Chaining obstacles, Field book, Scale; Compass surveying: - Bearings, WCB and RB system, Fore Bearing and Back Bearing, Local attraction and correction; Levelling: -Terms and definitions, Types of levels, Reduction of levels, Contouring; Area and volume measurement; Use of digital planimeter.
- c) **MECHANICS OF MATERIALS:** Moment of Inertia and Radius of Gyration; Section Modulus; M.I. of standard and built-up sections; Stress and Strain; Hooke's law; Elastic constants; Temperature stress and strain; Composite sections under axial loading; Shear Force and Bending Moment; SFD and BMD for: Cantilever beams, Simply supported beams; Bending stress and shear stress in beams; Columns: Short and long columns, Buckling and crushing load, Euler's and Rankine's concept.
- d) **BUILDING CONSTRUCTION:** Components of a building and their functions; Foundation: shallow and deep foundations; Masonry construction; Damp proofing; Doors, windows, lintels, arches; Floors, roofs and staircases; Finishing works; Building services.
- e) **CONCRETE TECHNOLOGY:** Ingredients of concrete and their properties; Water-cement ratio; Workability of concrete; Mix design concept; Fresh and hardened concrete properties; Segregation and bleeding; Compaction and curing; Special concretes (basic introduction); Admixtures and additives.
- f) **GEOTECHNICAL ENGINEERING:** Origin and formation of soils; Soil classification; Index properties of soil; Water Content, Void Ratio, Porosity, Degree of Saturation; Permeability of soil; Compaction and consolidation; Shear strength of soil; Earth pressure; Bearing capacity of soil; Site investigation and sampling; Foundation soil suitability; Soil stabilization methods.
- g) **HYDRAULICS:** Properties of fluids; Pressure and its measurement; Flow through pipes; Bernoulli's theorem; Hydraulic gradient and energy gradient; Laminar and turbulent flow; Head loss in pipes; Flow measurement devices; Open channel flow.
- h) **ADVANCED SURVEYING:** Theodolite surveying; Tacheometric surveying; Curve setting; Plane table surveying; Total Station and modern surveying methods; Introduction to GPS/GNSS and remote methods.
- i) **BUILDING PLANNING & DRAWING:** Types of lines-visible lines, Centre line, Hidden line, Section line, Dimension line, Extension line, Pointers, Arrowhead, or dots. Appropriate size of lettering and numerals for titles, Sub-titles, Notes, and dimensions. Types of scale- Monumental, Intimate, Criteria for Proper Selection of

scale for various types of drawing. Sizes of various standard papers/sheets. Principles of planning for Residential and Public building- Aspect, Prospect, Orientation, Grouping, Privacy, Elegance, Flexibility, Circulation, Furniture requirements, Sanitation, Economy. Space requirement and norms for minimum dimension of different units in the residential and public buildings as per IS 962. Rules and byelaws of sanctioning authorities for construction work. Plot area built up area, Super built-up area, Plinth area, Carpet area, floor area and FAR (Floor Area Ratio).

- j) **TRANSPORTATION ENGINEERING:** Highway development and classification; Highway Alignment; Road cross-sections and geometric design basics; Traffic engineering basics; Traffic signals and design; Traffic signs and road markings; Pavement types: Flexible and rigid; Pavement components and functions.
- k) **DESIGN OF RCC STRUCTURES:** Limit State Method; Design philosophies; Reinforcement materials; Loading on structures as per IS 875; Shear, bond and development length; Singly reinforced beam; Doubly reinforced beam; One-way slab; Two-way slab; Axially loaded columns; Basic IS 456:2000 provisions.
- l) **ESTIMATING & COSTING:** Meaning and purpose of estimating and costing; Types of estimates: -Approximate, Detailed, Revised, Supplementary; Measurement rules as per BIS 1200; Centre line and long wall-short wall method; Detailed estimate of: - Building, Septic tank, Road work; Rate analysis; Tendering and contracts; Schedule of Rates; Administrative Approval and Technical Sanction; Construction management basics: -Planning, Scheduling, Labour and material control, Site organization, Time-cost concepts.
- m) **WATER RESOURCE ENGINEERING:** Hydrological cycle; Rain gauges and rainfall measurement; Average rainfall methods; Runoff and factors affecting runoff; Crop water requirement; Duty, Delta, CCA, GCA, irrigation intensity; Reservoir planning and silting; Dams and spillways; Lift irrigation; Drip and sprinkler irrigation; Weirs, barrages, canals; Water logging and its control.
- n) **PUBLIC HEALTH ENGINEERING:** Water supply system components; Sources of water and treatment basics; Distribution systems; Sewerage system basics; House drainage and sanitary fittings.
- o) **DESIGN OF STEEL STRUCTURES:** Structural steel sections; Tension and compression members; Bolted and welded connections; Roof trusses; Steel column and beam concepts.
- p) **EARTHQUAKE RESISTANT BUILDING DESIGN:** Causes and effects of earthquakes; Seismic zoning of India; Earthquake-resistant features in buildings; Ductile detailing concept; IS 1893, IS 13920, IS 4326; Retrofitting measures.
- q) **PAVEMENT DESIGN AND MAINTENANCE:** Types of pavements – Flexible, Rigid and Semi Rigid; Factors affecting pavement design-design; Design overview of Flexible and Concrete pavement: Methods of Pavement evaluation; Pavement Maintenance.
- r) **PRECAST/PRESTRESSED CONCRETE:** Precast concrete elements; Prefabricated building components; Prestressing concept; Pre-tensioning and post-tensioning; Losses of prestress; Applications of prestressed concrete.
- s) **GREEN BUILDINGS:** Definition and Principles; Energy efficiency; Water efficiency; Environmental Impact Assessment; Green materials; LEED / rating concept.
- t) **CONSTRUCTION MANAGEMENT:** Construction Management; Planning; Scheduling; Labour and material control; Site organization; Time-cost concepts.
- u) **SOLID WASTE MANAGEMENT:** Collection; Segregation; Transportation; Disposal.

- v) **RAILWAYS, BRIDGES AND TUNNELS: RAILWAYS:-**Introduction to Indian Railways
• Railways surveys: Factors influencing the railways route, brief description of various types of railway survey • Classification of permanent way describing its component part • Rail Gauge: Definition, types, practice in India • Rail – types of rails • Rail Fastening: Rail joints, types of rail joints, fastening for rails, Fish plates, spikes bearing plates • Sleepers: Functions of sleepers, types of sleepers, requirements of an ideal material of Sleepers. • Ballast: Function of ballast, requirements of an ideal material of ballast • Crossing and signalling: Brief description regarding different types of crossing/signalling • Maintenance of track: Necessity, track fixtures; maintenance and boxing of ballast, maintenance gauges, tools • Drains, methods of construction. **BRIDGES:-** Introduction • Bridge–its function, component parts, suitability, difference between bridge and culvert • Classification of Bridges: According to life–permanent and temporary; According to deck level–Deck, through and semi-through; According to material–timber, masonry, steel, RCC, pre-stressed • IRC classification • Bridge Foundations: Open foundation, Pile foundation, Well foundation • Piers, Abutments and Wing walls • Piers–definition, parts; types–solid (masonry and RCC) • Abutment and wing walls–definition, types of abutment (straight and tee), abutment with wing walls (straight, splayed, return and curved) • Bridge bearings, purpose of bearing, types of bearing–fixed plate, rocker and roller • Maintenance of Bridges • Inspection of bridges • Routine maintenance. **TUNNELS:-** • Definition and necessity of tunnels • Typical section of tunnels for a national highway, single and double broad gauge railway track • Ventilation–necessity and methods of ventilation: by blowing, exhaust and combination of blowing and exhaust • Drainage method of draining water in tunnels • Lighting in tunnels & lining of tunnels.

2. **GENERAL AWARENESS**

- (a) **General knowledge**: General Knowledge including General knowledge of Himachal Pradesh.
(b) **Current Affairs** .
(c) **Everyday Science** .
(d) **Logical Reasoning** .
(e) **Social Science** (10th standard).
(f) **General English** (10th standard).
(g) **General Hindi** (10th standard).